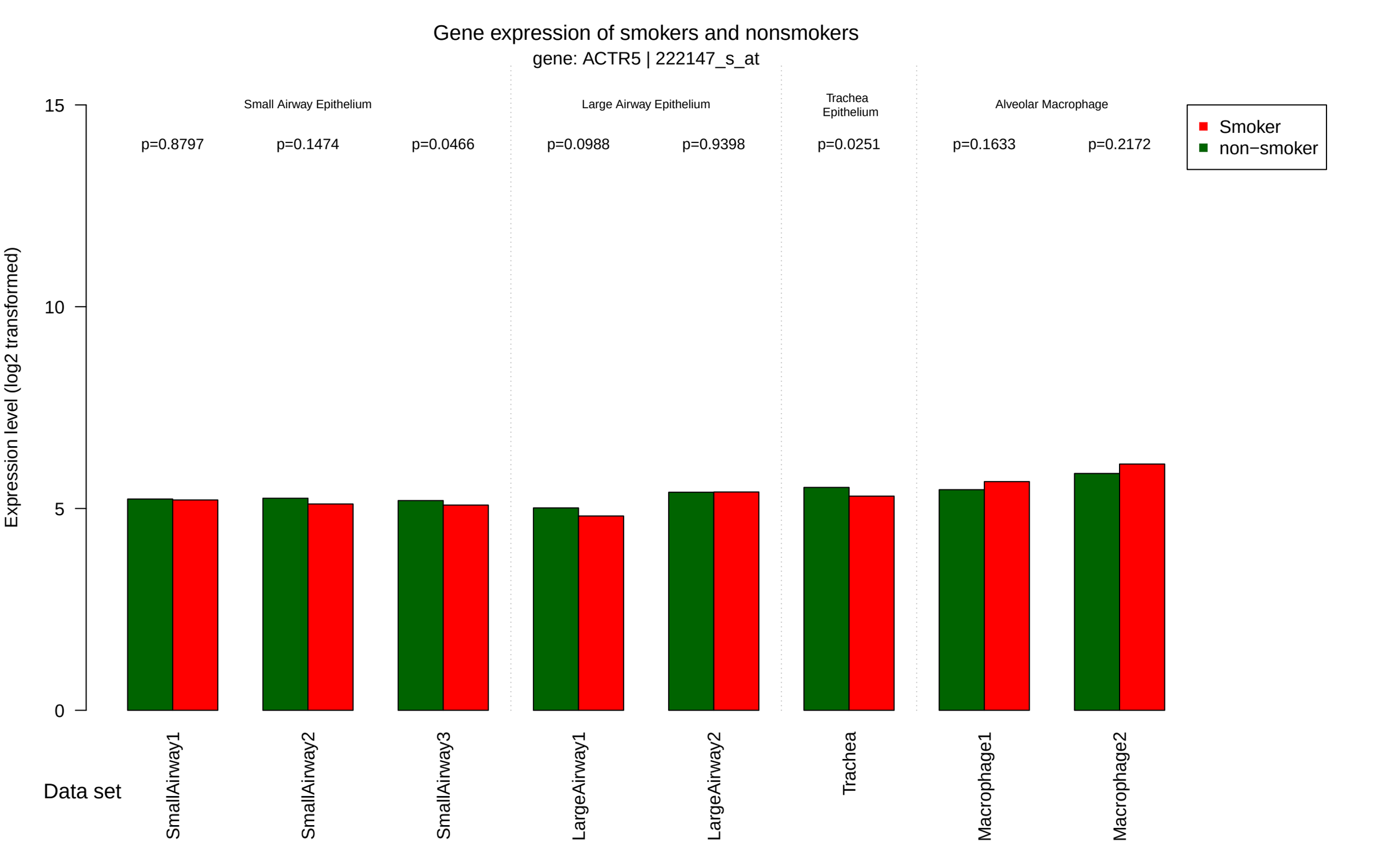
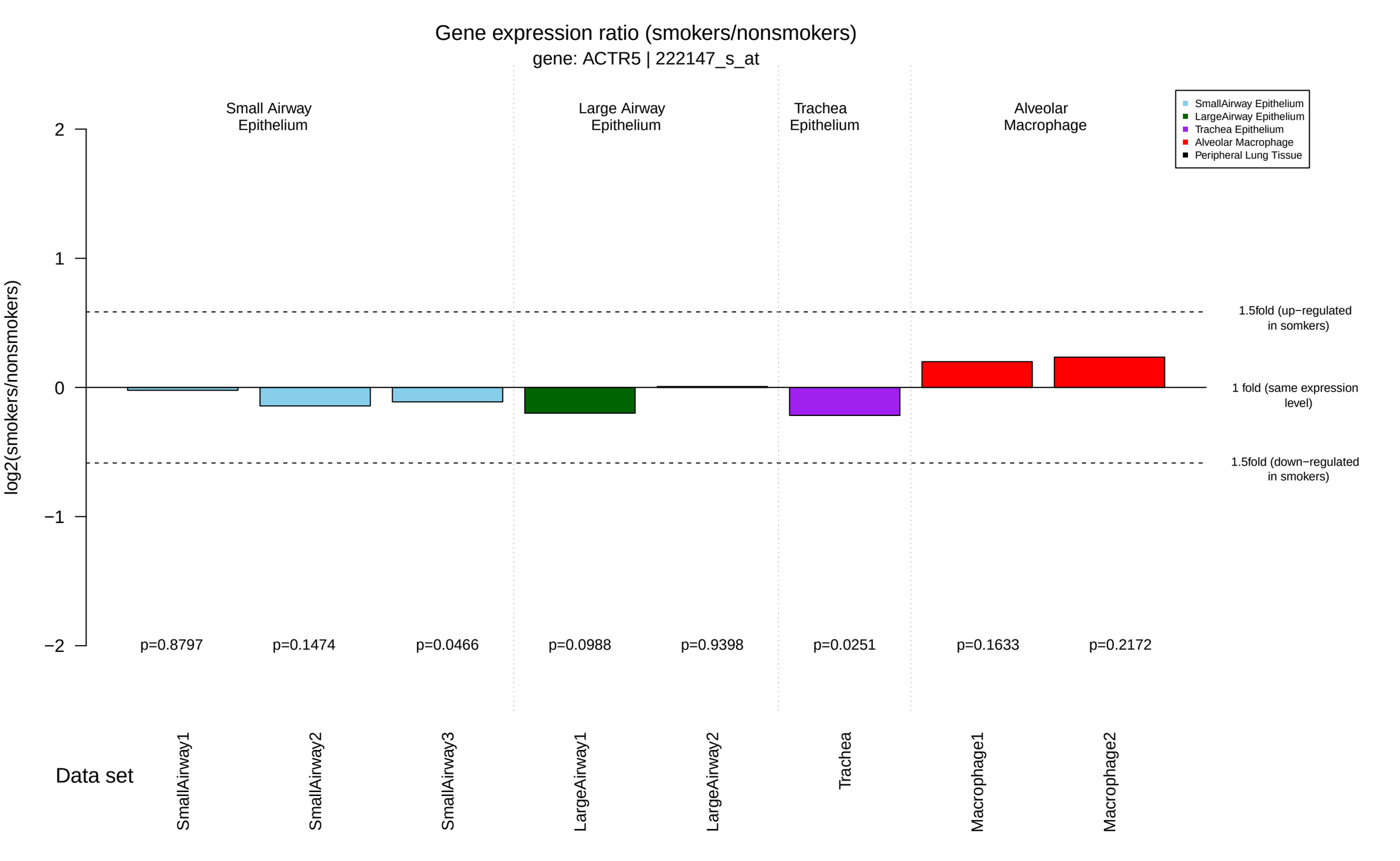
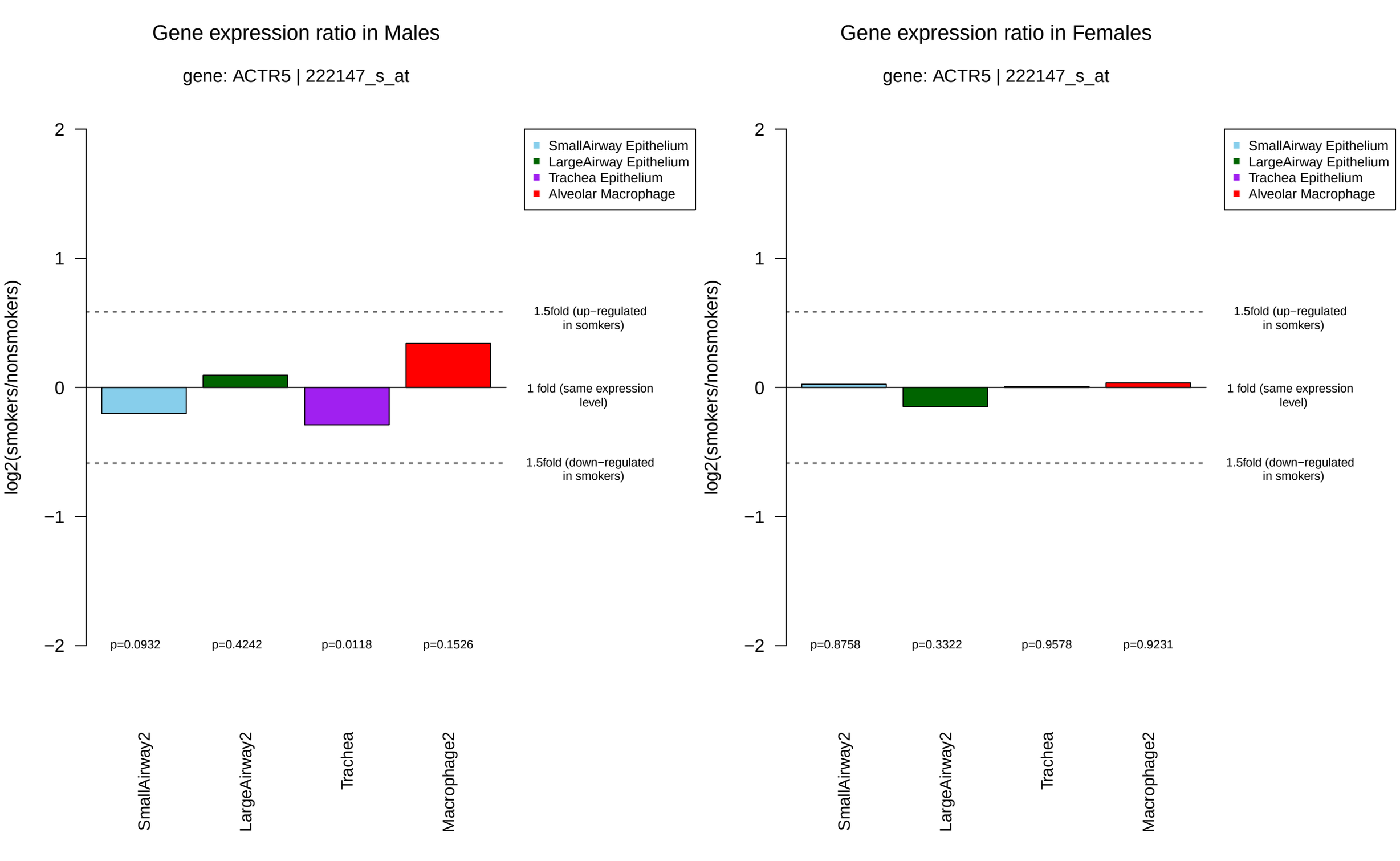
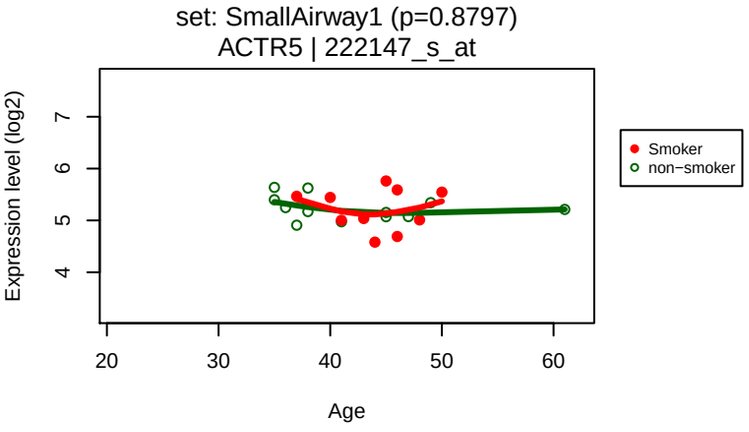
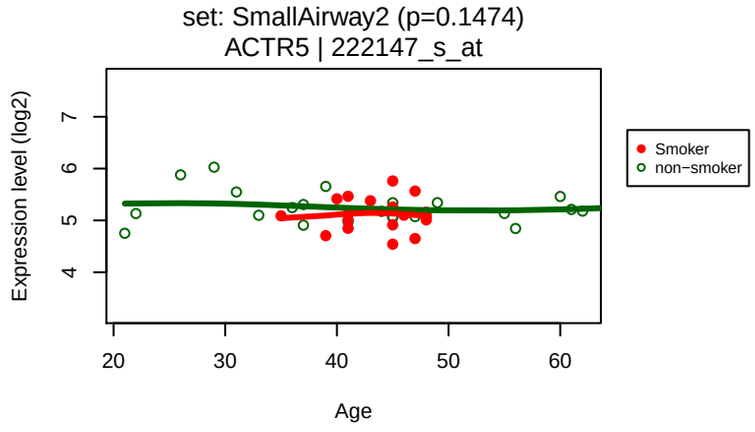
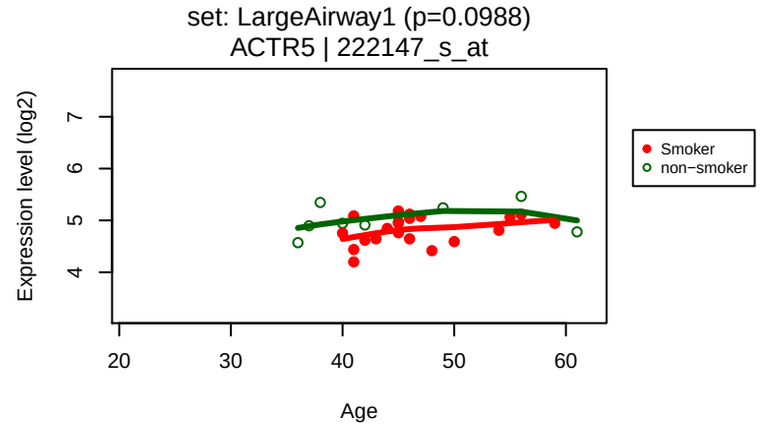

Gene expression of smokers and nonsmokers
gene: ACTR5 | 222147_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8797
p=0.1474
p=0.0466
p=0.0988
p=0.9398
p=0.0251
p=0.1633
p=0.2172
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

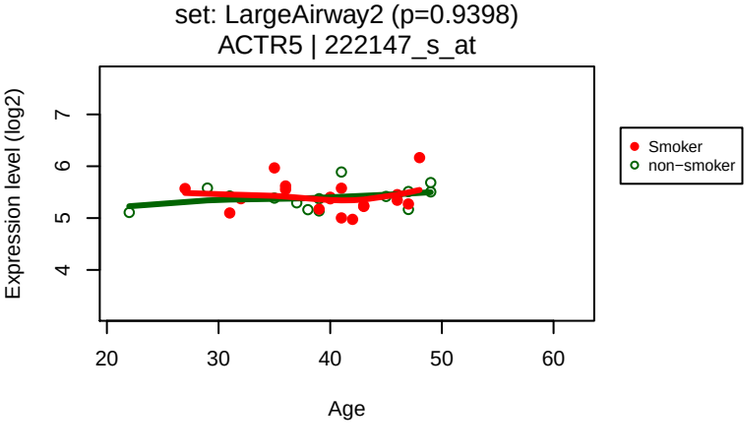

Gene expression ratio (smokers/nonsmokers)
gene: ACTR5 | 222147_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8797
p=0.1474
p=0.0466
p=0.0988
p=0.9398
p=0.0251
p=0.1633
p=0.2172
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

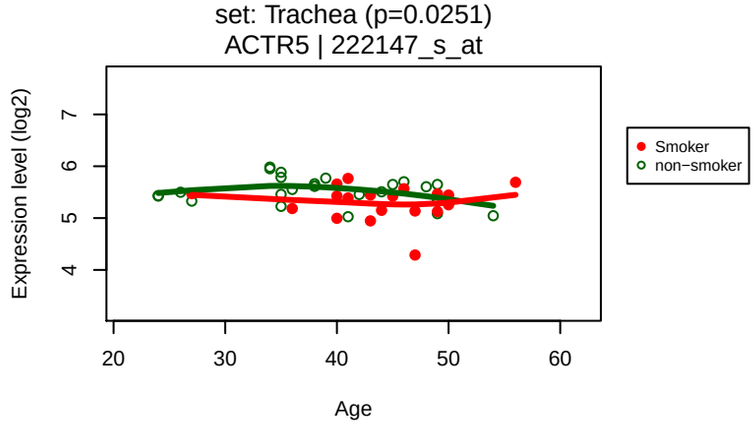

Gene expression ratio in Males
gene: ACTR5 | 222147_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0932
p=0.4242
p=0.0118
p=0.1526
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR5 | 222147_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8758
p=0.3322
p=0.9578
p=0.9231
SmallAirway2
LargeAirway2
Trachea
Macrophage2

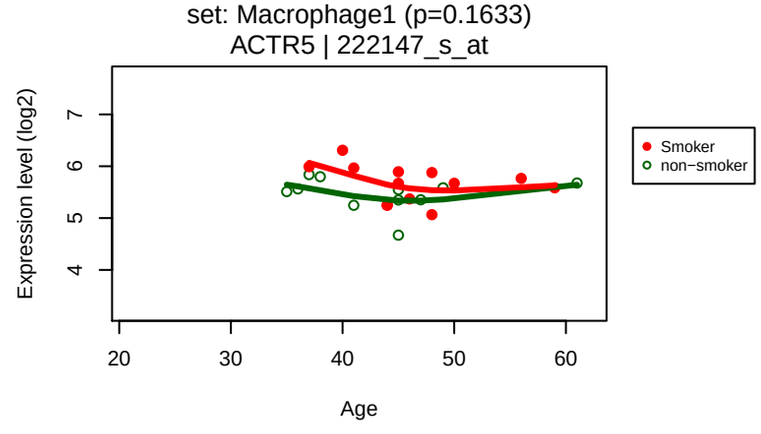

set: SmallAirway1 (p=0.8797)
ACTR5 | 222147_s_at
Expression level (log2)
Age
Smoker
non−smoker

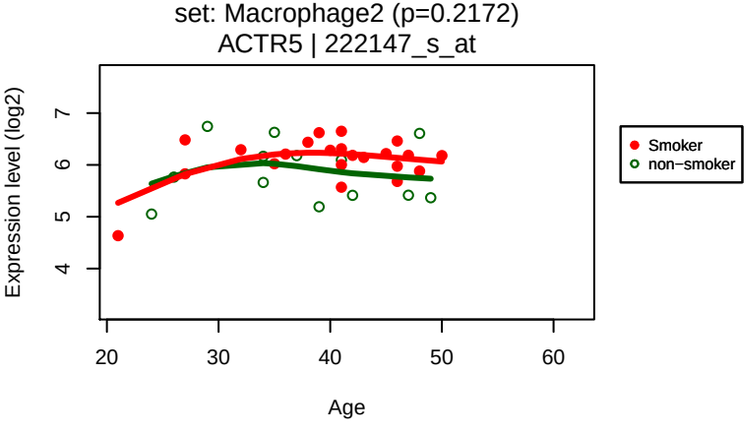

set: SmallAirway2 (p=0.1474)
ACTR5 | 222147_s_at
Expression level (log2)
Age
Smoker
non−smoker

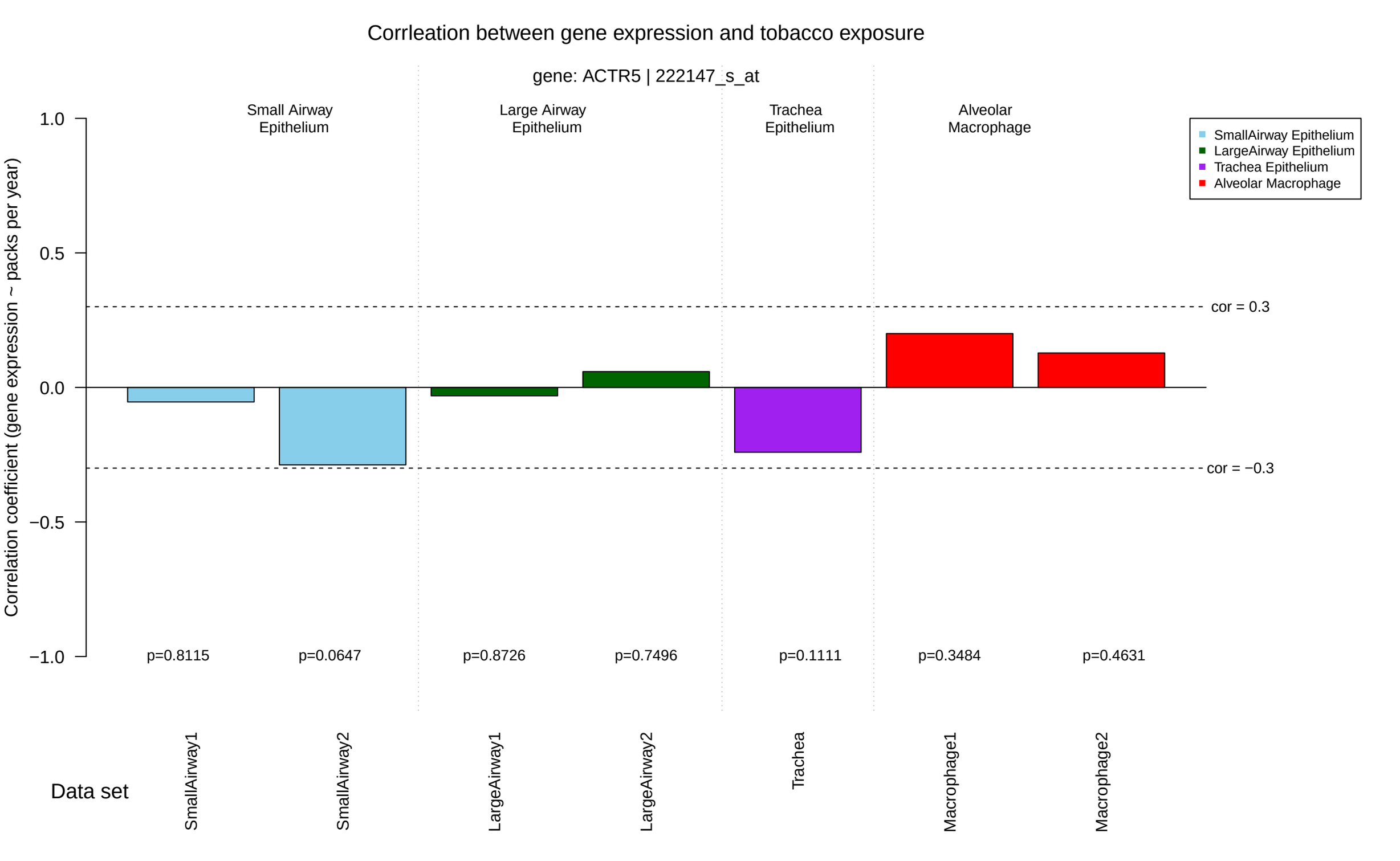

set: LargeAirway1 (p=0.0988)
ACTR5 | 222147_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9398)
ACTR5 | 222147_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0251)
ACTR5 | 222147_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1633)
ACTR5 | 222147_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2172)
ACTR5 | 222147_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR5 | 222147_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8115
p=0.0647
p=0.8726
p=0.7496
p=0.1111
p=0.3484
p=0.4631
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2